

Ike Peng

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EDUCATION

Johns Hopkins University

Master of Science in Computer Science

Aug. 2026 – Dec. 2027 (Expected)

University of Chicago

Bachelor of Science in Computer Science and Statistics; GPA: 3.8/4.0

Sep. 2023 – Jun. 2026

EXPERIENCE

Intelligence Cubed (i³)

Software Engineer Intern

May 2026 – Aug. 2026

San Francisco, CA

- Architected an **AI coding platform** backend with **FastAPI**, **WebSockets**, **Redis**, and **PostgreSQL**, orchestrating persistent **Claude Code** sessions and real-time terminal state for **2,000+ developers** across concurrent workspaces.
- Built **agent execution infrastructure** on **AWS** with **Docker** sandboxing and automated VM provisioning, enabling workload orchestration, session recovery, resource scheduling, workspace isolation, and persistent storage.
- Designed repository-aware **RAG** and **multi-agent orchestration** with planner–implementer–reviewer loops, **tool calling**, and evaluator-guided retries, improving task completion by **10%** and reducing manual intervention by **30%**.

Roxxem

AI Engineer Intern

Jun. 2025 – Aug. 2025

Chicago, IL

- Built a real-time personalization service that transformed learner history and song metadata from **MongoDB** and **BigQuery** into difficulty and recommendation features for **10,000+ songs**.
- Productionized difficulty-model training and **FastAPI** inference with automated feature pipelines and model versioning, replacing LLM-based scoring and reducing latency from **2 s to sub-200 ms**.
- Built experimentation and **model-monitoring** pipelines for personalized predictions, tracking engagement and model drift as the learning product added **1,000+ paid users within 3 months**.

Kang Tao Technologies

Machine Learning Engineer Intern

Jul. 2024 – Jan. 2025

Beijing, China (Remote)

- Engineered a multimodal generation pipeline with **Stable Diffusion**, **LoRA**, and **ControlNet** to produce character-consistent image and video assets for a **30-minute AI-generated short film**.
- Automated preprocessing, captioning, **LoRA fine-tuning**, evaluation, and generation across **1,000+ images**, **30+ models**, and **20+ reusable workflows**.

PROJECTS

Forge — Self-Improving Coding Agent | *Python, PyTorch, Hugging Face, Docker, FastAPI*

- Built a **coding-agent runtime** with repository search, code editing, **tool calling**, terminal execution, and isolated **Docker** sandboxes; benchmarked it across **200+ repository-level software engineering tasks**.
- Developed a self-improvement and **post-training** loop that converted agent execution traces into **LoRA** fine-tuning data and benchmarked checkpoints on task success, latency, and token usage.

RESEARCH

Concept Incongruence: An Exploration of Time and Death in Role Playing | *NeurIPS 2025*

2025

- Built automated **LLM evaluation** and **probing pipelines** to measure factual accuracy and abstention behavior across leading language models under conflicting role-playing prompts.
- Analyzed **LLM hallucination** and temporal inconsistency when prompts conflicted with character timelines, identifying failure modes in role-playing behavior.

TECHNICAL SKILLS

Programming: Python, Java, C/C++, Rust, SQL

AI & Agent Systems: RAG, Embeddings, Tool Calling, Multi-Agent Systems, Agent Orchestration, LLM Evaluation, Transformers, Fine-tuning, LoRA/PEFT, Hugging Face

Backend & Data: FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Firebase

Systems & Cloud: AWS, GCP, Docker, Linux, Distributed Systems, Concurrency, AsyncIO, Multithreading, CI/CD, Git

ML & Data: PyTorch, scikit-learn, XGBoost, NumPy, Pandas, Model Evaluation, BigQuery, Stable Diffusion, ControlNet